

# Enhancing Primary Healthcare through Reliable Energy: Authorizing a Needs Assessment for Kenyan Clinics

**For: County health planning and finance officials**

Jurisdiction: Kenya; pilot county to be confirmed

Working draft. Confirm the pilot jurisdiction, responsible institutions and authority before submission.

## DECISION REQUESTED

Authorize a local needs assessment and costed options appraisal before any pilot procurement.

## KEY MESSAGES

1. Many primary healthcare clinics in Kenya lack reliable electricity, hindering essential services and patient care [7] [1] [8].
2. Clean energy solutions can provide reliable power, but local context and costed options are needed to inform investment [6].
3. A targeted needs assessment and options appraisal will provide data to make an informed decision on a pilot energy solution.

## Why this decision matters

This brief outlines a proposed approach to address unreliable energy access at public primary-care clinics in Kenya. Improving energy access is critical for delivering consistent healthcare services to affected populations, supporting public health infrastructure.

Reliable electricity is fundamental for primary healthcare, powering life-saving equipment and ensuring safe childbirth [8]. Many rural health facilities in Kenya face persistent energy gaps due to slow grid expansion and insufficient county-level funding for energy plans [7] [1]. Without consistent power, patient care suffers, and vital vaccines can spoil [8]. The proposed National Energy Policy 2025-2034 aims for universal electricity access by 2030, prioritizing renewable energy [3]. The Kenya Climate Change and Health Strategy 2024-2029 also calls for clean energy adoption in health facilities [2].

## Evidence and policy implications

**Persistent energy gaps in remote areas of Kenya are caused by slow national grid expansion, capacity constraints, insufficient county-level funding for energy plans, and a lack of reliable data and analytical tools [7] [1].**

Addressing energy reliability in primary healthcare requires targeted, data-driven planning at the local level, considering existing infrastructure limitations and funding challenges.

Consistent power is essential for powering life-saving equipment, ensuring safe childbirth, and operating critical devices such as oxygen concentrators, sterilization units, laboratory diagnostics, and vaccine refrigerators in health facilities [8]. Without it, patient care suffers, and vaccines can spoil [8].

Unreliable energy directly compromises patient safety, quality of care, and the efficacy of public health programs, necessitating urgent and dependable power solutions.

Solar photovoltaic (PV) and battery systems are proven to offer clean, reliable, quick, and cost-effective solutions for electrifying rural health facilities [6]. This aligns with existing policy frameworks emphasizing data-driven planning and local solutions [7] [5].

Renewable energy technologies, particularly solar, present a viable and policy-aligned pathway to improve energy access for rural clinics, but local assessment is crucial for optimal deployment.

### Compare the policy options

| Option and approach                                                                                                                                             | Benefits and trade-offs                                                                                                                                                                                                            | Feasibility                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Continue Current Energy Arrangements</b><br>Rely on existing grid connections (if available), diesel generators, or no electricity, with ad-hoc maintenance. | Benefits: No immediate new capital outlay or changes to current operational practices are required.<br>Trade-offs: Continued unreliable power, high operational costs for generators, and compromised patient care [8].            | Requires no new authorization or capacity, but fails to address core problem [7] [1].<br>[7] [1] [8] |
| <b>Public Ownership with Funded Maintenance</b><br>County procures, owns, and maintains clean energy systems (e.g., solar PV with batteries) for clinics.       | Benefits: Offers direct control over assets and potential for long-term cost savings with proper management.<br>Trade-offs: Requires upfront capital investment, dedicated maintenance budget, and technical capacity development. | Requires confirmed funding, technical expertise, and a robust maintenance framework.                 |
| <b>Competitively Procured Service Contract</b><br>Contract a private provider for energy services, covering uptime, maintenance, and fault response.            | Benefits: Transfers operational and maintenance risks, offers predictable costs, and guaranteed service levels.<br>Trade-offs: Requires robust contract management, potential for higher long-term costs, and less direct control. | Requires a clear procurement framework, legal authority for service contracts, and market interest.  |

### Recommended actions

#### 1. Authorize a comprehensive local needs assessment for selected primary-care clinics to identify current energy access, essential electrical loads, and existing infrastructure.

Lead: County Health Department, in collaboration with County Planning and Finance Officials | Proposed timing: Proposed: Within 3 months of authorization

Resources: Internal staff time, travel budget for site visits, and consultation with clinic staff and patient representatives.

Success measure: Completion of a detailed report outlining energy needs, current costs, and infrastructure gaps for target clinics.

## 2. Conduct a costed options appraisal comparing the whole-life costs and benefits of the three identified energy solutions (status-quo, public ownership, service contract) based on assessment findings.

Lead: County Planning and Finance Officials, supported by the County Health Department | Proposed timing: Proposed: Within 2 months following the needs assessment report.

Resources: Internal analytical capacity, access to financial modeling tools, and consultation with technical experts.

Success measure: Production of a comparative financial and operational analysis for each option, including fiscal and equity effects.

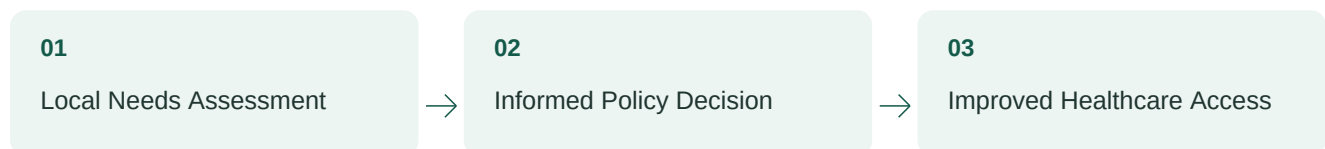
## 3. Present the needs assessment and costed options appraisal to relevant county decision-makers for review and to inform a decision on a potential pilot project.

Lead: County Planning and Finance Officials | Proposed timing: Proposed: Within 1 month of completing the options appraisal.

Resources: Preparation of presentation materials and scheduling of review meetings.

Success measure: Formal review and discussion by county decision-makers, leading to a decision on next steps for a pilot.

## Proposed pathway to impact



## Decision and immediate next steps

We request authorization to proceed with the local needs assessment and costed options appraisal. This will provide the necessary data to make an informed decision on a pilot energy solution for primary-care clinics. A decision point is proposed within six months.

## Evidence gaps and conditions

- The exact scope and scale of energy needs across all primary-care clinics in the pilot county remain to be fully quantified.
- The availability of specific funding mechanisms for capital investment or service contracts at the county level needs confirmation.
- The capacity of county teams to manage complex energy projects or service contracts requires further assessment.

## About the authors

Prepared by GSL demonstration team (illustrative example). Based on a strategy developed through Global Solutions Lab; the recommendations represent the team's proposal.

## References

### [1] [worldbank.org](https://openknowledge.worldbank.org/server/api/core/bitstreams/0b60fc4c-2cf0-598e-ac35-a25fd72e8507/content)

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### [2] [Launch of the Kenya Climate Change and Health Strategy - KEMRI](https://www.kemri.go.ke/launch-of-the-kenya-climate-change-and-health-strategy/)

<https://www.kemri.go.ke/launch-of-the-kenya-climate-change-and-health-strategy/>

**[3] Final Draft National Energy Policy 2025 November 2025.pdf - www.energy.go.ke**

[https://www.energy.go.ke/sites/default/files/Final%20Draft\\_National%20Energy%20Policy%202025\\_November%202025.pdf](https://www.energy.go.ke/sites/default/files/Final%20Draft_National%20Energy%20Policy%202025_November%202025.pdf)

**[5] nih.gov**

<https://pmc.ncbi.nlm.nih.gov/articles/PMC12851313/>

**[6] nih.gov**

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**[7] One County in Rural Kenya Is Using Clean Energy to Close its Electricity Gap | World Resources Institute**

<https://www.wri.org/insights/makueni-county-kenya-clean-energy-plan>

**[8] Rural Health Clinics Gain Reliable Power through Clean Energy; Data-driven and Inclusive Energy Planning Grows | World Resources Institute**

<https://www.wri.org/outcomes/rural-health-clinics-gain-reliable-power-through-clean-energy-data-driven-and-inclusive>